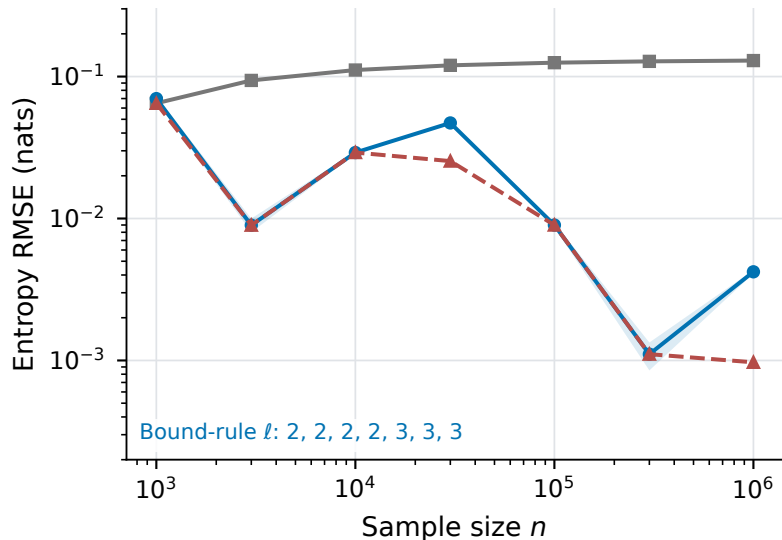


Bounded dependent density: error versus sample size ($d = 2$)

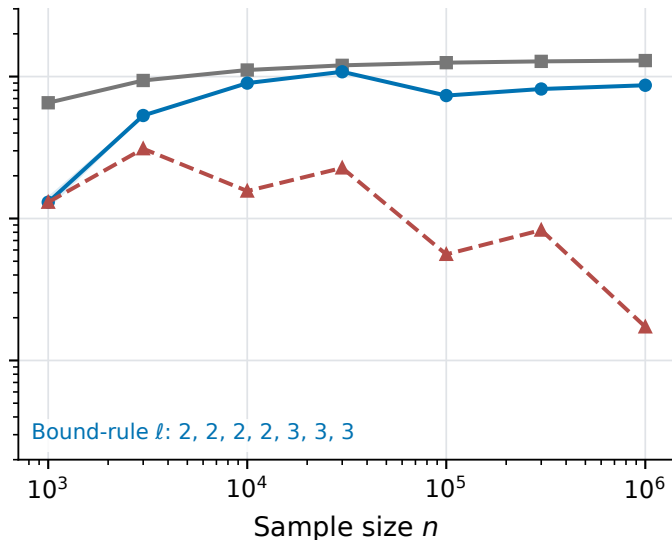
$f(x) = 1 + 0.7\cos(2\pi k(x_1 - x_2))$ on $[0, 1]^d$; true entropy = -0.131623 nats

—■— Fixed $\ell = 1$ —●— Bound rule ($\delta = 0.05$) -▲- Oracle reference

(a) Frequency $k = 1$



(b) Frequency $k = 2$



$n \leq 10^5$: 100 repeats; $n = 3 \times 10^5, 10^6$: 30 repeats. Shading: pointwise 95% bootstrap intervals.

Oracle: same-repeat minimum RMSE among fully covered candidates; optimistic reference, without a band.

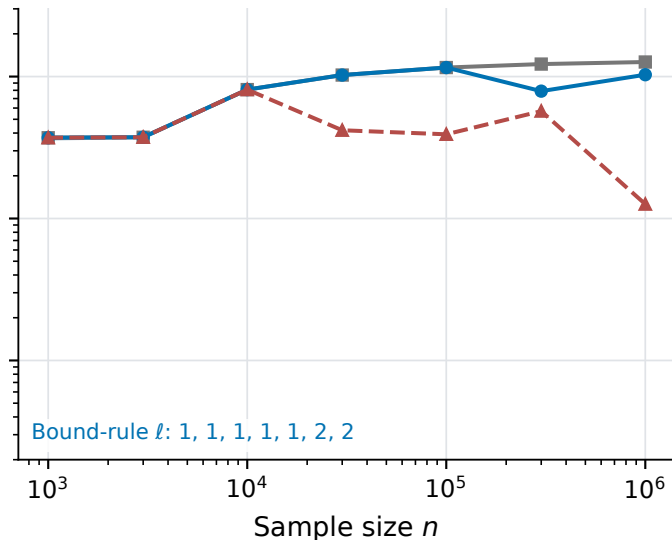
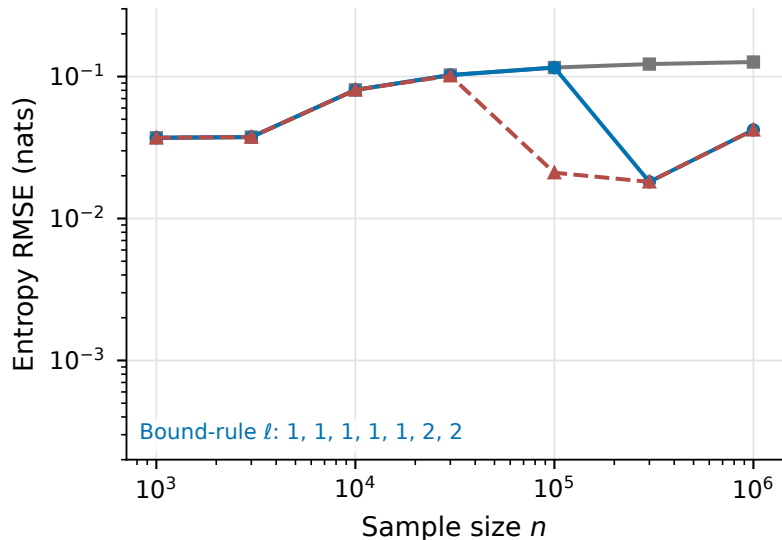
Bounded dependent density: error versus sample size ($d = 5$)

$f(x) = 1 + 0.7\cos(2\pi k(x_1 - x_2))$ on $[0, 1]^d$; true entropy = -0.131623 nats

—■— Fixed $\ell = 1$ —●— Bound rule ($\delta = 0.05$) - -▲- Oracle reference

(a) Frequency $k = 1$

(b) Frequency $k = 2$



$n \leq 10^5$: 100 repeats; $n = 3 \times 10^5, 10^6$: 30 repeats. Shading: pointwise 95% bootstrap intervals.

Oracle: same-repeat minimum RMSE among fully covered candidates; optimistic reference, without a band.